MIL-S-16829A(BuMed)
7 January 1966
SUPERSEDING
MIL-S-16829(BuMed)
15 November 1951

MILITARY SPECIFICATION

STAND, IRRIGATOR, STEEL, FOLDING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a folding irrigator stand suitable for use in field hospitals.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited).
TT-E-527	Enamel, Alkyd, Lusterless.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
MIL-P-52192	Primer Coating, Epoxy.

FSC 6530

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STANDARDS

Federal

FED-STD-66 Steel; Chemical Composition and Hardenability.
FED-TEST METHOD STD 151 Metals; Test Methods.
FED-STD-595 Colors

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Medical Supply Center

20519 Stand, Irrigator, Steel, Folding

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1), prior to the start of production of the contract quantity, one preproduction sample of the irrigator stand shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted,together with the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Material.

3.2.1 Base and uprights. The base and braces shall be AISI 1018 steel, and the tubular uprights shall be AISI 1010 steel. Both AISI 1018 steel, and AISI 1010 steel shall be in accordance with FED-STD-66 when tested as specified in 4.4.1.

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3.2.2 Brace collars. Brace collars shall be cast brass or bronze.

3.3 Construction. The irrigator stand shall be of the collapsible type consisting of a folding tripod base with base upright supporting two telescoping extension uprights. Construction shall be in accordance with drawing 20519. The three joints where the braces meet the legs shall be provided with No. 14 standard brass washers.

3.3.1 Tripod base. Tripod base and braces shall be made of rectangular steel bars (flats) 1/2-inch wide by 1/8-inch thick. It shall consist of three legs attached at their upper ends by riveted pivot joints to a sliding adjustable bronze collar clamp on 120° centers. The clamp shall encircle the base upright and shall be provided with a winged head set screw. The tripod shall be provided with three braces, which shall be attached peripherally to each leg by a riveted pivot joint as shown on drawing 20519. The braces shall be similarly attached centrally on 120° centers to a stationary bronze collar clamp, welded to the bottom of the base upright (see 3.3.2). Construction shall be such, that by loosening the set screw on the sliding adjustable collar, the tripod shall fold with legs and braces in contact with the base upright.

3.3.2 Base upright. Base upright shall be made of steel tubing 5/8-inch in outside diameter, having a wall thickness of 0.035 inch (No. 20 Birmingham Wire (B.W.) gage). Upright shall be securely welded to the brace collar and shall pass through the sliding adjustable leg collar. At the upper end of the upright, there shall be securely attached a steel collar, 3/8-inch wide, through which shall be threaded a 3/16-inch diameter winged setscrew. The setscrew shall be of sufficient length to hold the tubular upright securely in place.

3.3.3 Telescoping extension uprights. There shall be two telescoping extension uprights; a lower tubular and an upper rod. The lower tubular extension upright shall telescope within the base upright while the upper rod extension shall telescope within the lower tubular member. Construction and dimensions of the uprights shall be such as to insure ease in telescoping, with a minimum of sticking, bending, or side to side play.

3.3.3.1 Tubular upright. Tubular upright shall be 1/2-inch in diameter, 0.035 inch thick (No. 20 B.W. gage). The upper end shall be provided with a steel collar, 3/8-inch wide, through which shall be threaded a 3/16-inch diameter winged setscrew. The setscrew shall be of sufficient length to hold the rod extension upright firmly in place.

3.3.3.2 Rod extension upright. Rod extension upright shall be 3/8-inch diameter and of steel rod stock. The upper end shall be formed with an intergral hook of design and dimensions as shown on drawing 20519.

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3.4 Dimensions. Dimensions shall be as shown on drawing 20519. When all uprights are telescoped and tripod is folded, overall dimensions shall not exceed 29-1/2 inches in length and 2-1/2 inches in width.

3.5 Finish. All parts and surfaces shall be cadmium plated to comply with class 2, type III plating of QQ-P-416. All parts shall be coated with primer specified in MIL-P-52192. All parts shall then be top coated within 4 hours with lustreless enamel, in accordance with TT-E-527, color X34087 of FED-STD-595. The threaded inserts shall be coated with zinc chromate primer specified in MIL-P-8585 prior to insertion to castings.

3.6 Identification marking. Each irrigator stand shall have a metal nameplate securely attached and permanently marked with the following:

Manufacturer's name or registered trademark
Contract or purchase order number
Federal stock number
Letters U.S.

3.7 Workmanship. Workmanship shall be first class throughout. Irrigator stand shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105

	Inspection level	AQL (defect per hundred units)
For visual examination Major defects	II	1.0
Total (major and minor combined) defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

4.3 Examination. Tne stands shall be examined to determine compliance with all requirements contained in this specification.

	Categories and defects *
Major	
101	Stand not complete with all components.
102	Unit not free of burrs, rough surfaces, and sharp edges.
103	Stand does not rest in a vertical steady position.
104	Any component does not function as specified.
Minor	
201	Any weld not continuous, not clean, not free of pitting, splatter and discoloration.
202	Unit not clean; not free of dirt, oil, and grease.
203	Finish not as specified.
204	Identification marking not present, properly located, legible, correct, or permanently affixed.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The stands shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

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4.4 Tests. Tests shall be conducted, to determine compliance with specification requirements.

4.4.1 Chemical composition. Test for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Fed. Test Method Std. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One stand, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not Applicable	12 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.2.1 and 5.2.2).
(c) Where preproduction sample shall be sent (see 3.1).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

CATEGORY N

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